Boats and Streams-II

1. A boat running upstream takes 8 hours 48 minutes to cover a certain distance, while it takes 4 hours to cover the same distance running downstream. What is the ratio between the speed of the boat and speed of the water current respectively?
2. A man takes thrice as long to row a distance against the stream as to row the same distance in favor of the stream. The ratio of the speed of the boat (in still water) and the stream is:
3. How long will it take to row 30 km upstream if one can row 10 km in 10 minutes in still water and the same distance in 8 minutes with the stream?
4. A man can row at 5 kmph in still water. If the velocity of current is 1 kmph and it takes him 1 hour to row to a place and come back, how far is the place?
5. A boat travels from point P to Q, a distance of 12 km. From P it travels 4 km downstream in 15 minutes and the remaining 8 km upstream to reach Q. If the downstream speed is twice as high as the upstream speed, what is the average speed of the boat for the journey from P to Q?
6. The speed of a boat in still water is 15 km/hr. It can go 30 km upstream and return downstream to the original point in 4 hours 30 minutes. Find the speed of the stream.
7. A boat running upstream takes 420 minutes to cover a certain distance, while it takes 240 minutes to cover the same distance running down stream. What is the ratio between the speed of the boat and speed of the water current, respectively?
8. Sanju in still water has recorded a row speed of 14 km/h. He takes thrice the time of rowing down the river to row up the river. What is the rate of this stream?
9. Consider a boat which moves at the speed of 6 km/hr. If the water runs at the speed of about 4 km/hr, then the boat requires 3 hours to reach a certain place and return. Calculate the distance between that place & boat's initial position.
10. Speed of a boat in still water is 18 km/hr and the speed of the stream is 3 km/hr. A man sails a distance of 105 km and comes back to the starting point. The total time taken is.
11. A man rows to a place 48 km distant and come back in 14 hours. He finds that he can row 4 km with the stream in the same time as 3 km against the stream. The rate of the stream is:
12. A man has to go from a port to an island and return. He can row a boat with a speed of 7 km/hr in still water. The speed of the stream is 2 km/hr. If he takes 56 minutes to complete the round trip, find the distance between the port and the island.

PRACTICE QUESTIONS:

1. A motorboat can travel at 5 km/hr in still water. It travelled 90 km downstream in a river and then returned, taking altogether 100 hours. Find the rate of flow of the river.
2. Two boats A and B start simultaneously from opposite ends of a 60 km river. Boat A moves downstream, and Boat B moves upstream. If A's speed in still water is12 km/hr, B's speed in still water is 8 km/hr, stream speed is 4 km/hr, which boat reaches the other end first, and by how much time?
3. Speed of a boat in standing water is 9 kmph and the speed of the stream is 3 kmph. A man rows to a place at a distance of 84 km and comes back to the starting point. The total time taken by him is:
4. A man rows to a place 96 km away and comes back in 28 hours. He finds that he can row 8 km with the stream in the same time as 6 km against the stream. The rate of the stream is.

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